Use of alternative data sources in CPI compilation. Bilateral and multilateral indices

UN-CEBD Task Team
on Scanner Data

Learning Objectives

By the end of the session, participants will:

- Be aware of the commonly used methods to compile elementary indices for the CPI with alternative data sources and,
- Understand the key methodological decisions, such as the choice between bilateral or multilateral indices.

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Contents

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Traditional index compilation

- Product categories and retailers are selected to represent consumer spending in each location.
- Within these categories, a sample of homogeneous products is chosen, defined by predetermined characteristics.
- Price changes for this sample, sometimes stratified by location or retailer type, are aggregated to form an elementary aggregate price index.
- Elementary aggregates are then weighted and combined to produce higher-level price indices.

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Elementary aggregate indices

- Elementary aggregates are the lowest-level indices, typically grouped by COICOP classification — but they can also be stratified by region, retailer, or retailer type.
- They form the building blocks of the CPI.
- With new data sources, several methods are available for calculating elementary aggregates.
- The choice of method depends on the NSO’s strategy and the properties of the data.

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Challenges and opportunities for index calculations with new data sources

Opportunities

- New data allow for enrichment of samples or even a universe of products to be included
- Availability of quantities (in transaction data) allow use of explicit weighting of products

Challenges

- Traditional “bilateral” indices are not well equipped to handle a dynamic universe of products observed in new data
- To make best use of the data, more complex “multilateral” index formulas are being proposed and adopted as the best approach, though such formulas are more intricate to understand and implement

Static and dynamic approaches

STATIC APPOACH

Product	Dec	Jan	Feb	Mar
Product 1	2.97	2.96	2.93	3.03
Product 2 ↓ Product 4	3.64	3.5	3.36	3.37

Static approach uses samples

- missing prices are imputed or replaced with new product.

The sample ideally consists of most sold representative products to keep costs low and the collection/quality assurance manageable

The number of products/rows is stable throughout the period

DYNAMIC APPROACH

Product	Dec	Jan	Feb	Mar
Product 1	2.97	2.96	2.93	3.03
Product 2	3.64	3.5	3.36	
Product 3	6.75	6.71	6.67	6.73
Product 4			3.37	3.37
Product 5	4.50	4.45		4.48
Product 6			5.39	5.16

Dynamic approach uses a census of all classified and verified transactions

Products that match in two or more periods enter calculation

The number of products/rows can be variable throughout the period

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Introduction to bilateral and multilateral indices

Bilaterals: only the data of the considered couple of periods is used in the compilation of the index

Base period   Comparison period

1 2 3 4 5 6 7 8 9 10 11 12 13

Multilaterals: data of the whole window of periods is considered to compile the index between the base and comparison periods

Base period   Comparison period

1 2 3 4 5 6 7 8 9 10 11 12 13

Whole windown of months considered

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Bilateral methods

- Bilateral methods measure price change between a base period and a current period.
- A direct bilateral index is where the base period is fixed for a measurement period and the price index chained periodically (e.g. annually).
- A chained bilateral index is where the base price moves throughout the measurement period and chained frequently (e.g. monthly).
- There are both weighted and unweighted bilateral indices available

Bilateral methods

	Static approach	Dynamic approach
No weights available	Direct: Jevons, Dutot	Chained: Jevons, Dutot
Weights available	Direct: Laspeyres (base period weights) Paasche (current period weights) Fisher, Tōrnqvist, Walsh (superlative – use base and current period weights)	Chained: Laspeyres, Paasche, Fisher, Tōrnqvist, Walsh ⚠ Prone to drift – use with caution

Multilateral methods

Multilateral index methods have traditionally been used for comparisons of price levels across countries or regions, but have recently become favored for processing large, dynamic data because these methods:

- Can incorporate all products in the dataset;
- Allow the use of time-varying product level weights;
- Continuous inclusion of new products, and
- Preserve transitivity in the index series.

Multilateral methods

The most commonly used multilateral methods are:

- The Geary-Khamis (GK)
  - an extension of a Unit Value method with a quality correction.
- The GEKS (Gini-Éltető-Köves-Szulc) method
  - transforms many underlying bilateral indices between all periods into one transitive index.
- The Time Product Dummy method
  - Estimates both price and quality changes using least squares regression.

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Multilateral methods

Choices must be made when using multilateral methods, such as:

- The multilateral index formula to be applied,
- The window length over which the method is applied,
- The splicing method used, and/or
- The aggregation structure and production system for integrating diverse data sources in the compilation.

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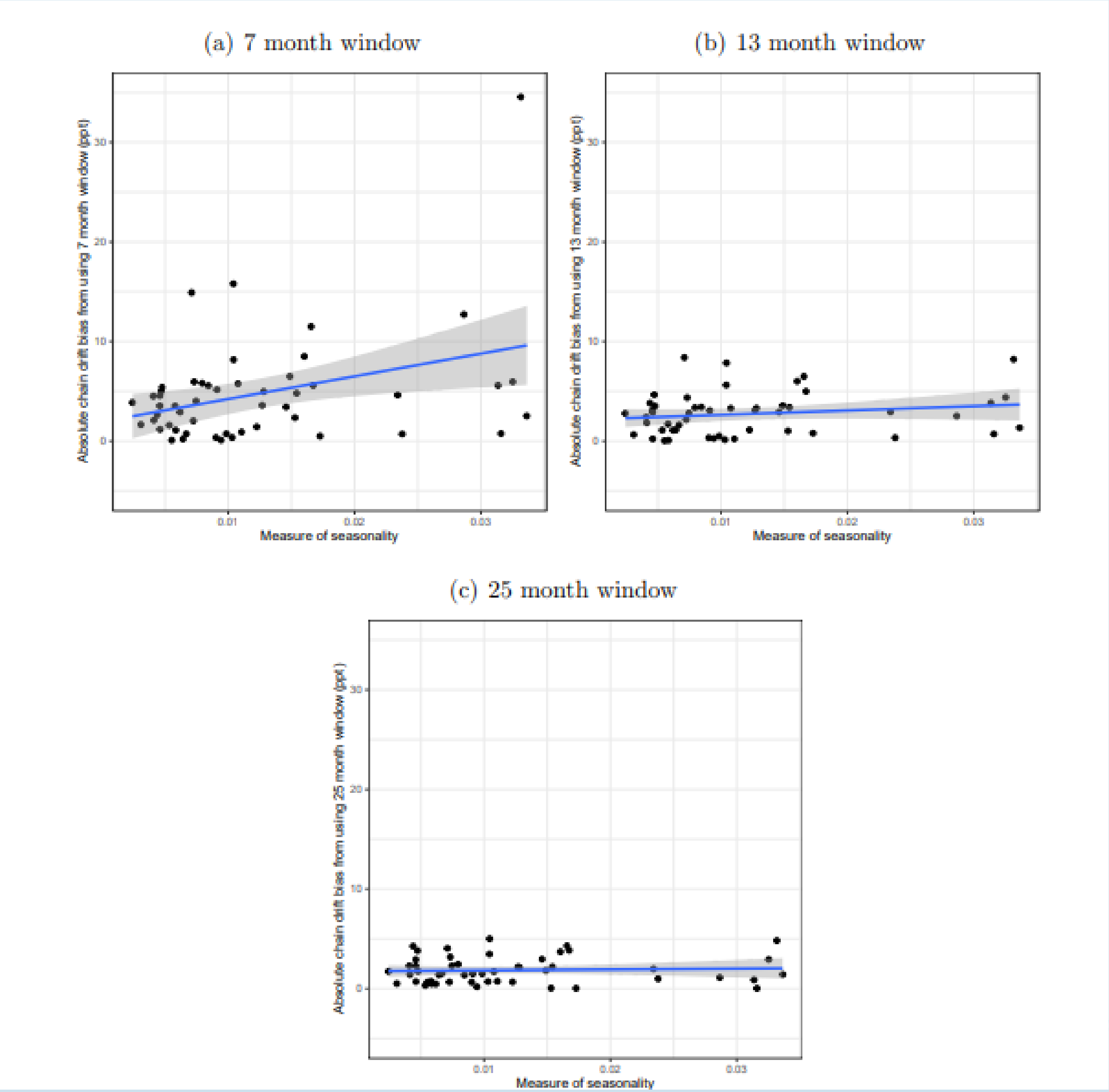
Window length

Window length	Advantages	Limitations	Best for
Short (<=13 months)	• Responsive to current price changes • Handles short product life cycles • Easier computationally	• May produce unstable results if few overlapping products • More susceptible to chain drift • Poor for strongly seasonal goods	• Fast-moving consumer electronics • Fashion items
Long (>=25 months)	• Captures two or more seasonal periods – more stable • Reduces volatility and drift	• Past data influences current estimates • Less responsive to market shifts	• Seasonal products • Slower-turnover goods

Seasonality and window lengths



Adjusting the time window

Rolling time windows	Expanding time windows
Each month, the time window is …	
…shifted forward by one month.	...extended by one month. After one year, it could be reset to its initial length.
Advantages	
Window length is constant	Does not need long back data

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Adjusting the time window example

Rolling window:

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Index period 4						
Index period 5						
Index period 6						

Expanding window:

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Index period 2						
Index period 3						
Index period 4						
Index period 5						
Index period 6						

Hybrid approach:

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Index period 2						
Index period 3						
Index period 4						
Index period 5						
Index period 6						

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Why do we need extension methods?

When using either a rolling or expanding window, a new time window is produced every month, producing a new index series.

To avoid revisions, extension methods must be used to link the latest result to the previous, ensuring the published index remains fixed.

Importantly, we need to ensure this doesn't result in ***drift.***

Splicing methods

The splicing of two series operates via a link period. There are different options of link period, for example:

Movement splice: The period t-1 is used as the link period

Window splice: The period t-T+1 is used as the link period

Half splice: The period t-(T+1)/2+1 is used as the link period

Mean splice: All the overlap periods are used as link periods

Fixed base: The previous December is used as the link period

Notation:

The current month - **t**

The window length – **T**

The link period can be from the recalculated or the published indices:

- link the multilateral index compiled in period t with the multilateral index compiled in period t – 1;
- link the multilateral index compiled in period t with the published index in period t – 1.

Useful notes on the splicing methods

Movement splice (link period: t-1)

- easily understood and explained
- consistent with the month-on-month movements obtained with the multilateral index
- chain drift cannot be excluded

Window splice (link period: t-T+1)

Half splice (link period: t-(T+1)/2+1)

- may have chain drift when linking onto previously calculated series
- alternative would be to link onto previously published series

Mean splice (link period: All the overlap periods)

- more robust alternative as it is based on all possible links.
- chain drift of movement or window splice can propagate to the mean splice

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What drives method choice?

Choosing a family of methods:

- Data availability
- Data quality
- Approach preference
- Revision policy

Narrowing to a specific method:

- Computational and staff capacity
- Theoretical properties
- Empirical performance
- Interpretability and explainability

Choosing a family of methods

	Static approach	Dynamic approach
Unweighted	Fixed base unweighted bilateral (e.g. fixed base Jevons)	Chained unweighted bilateral (e.g. chained Jevons) or multilateral (e.g. GEKS-Jevons)
Weighted	Fixed base weighted bilateral (e.g. Fixed base Tōrnqvist)	Weighted multilateral (e.g. Geary Khamis, GEKS-Tōrnqvist, Time Product Dummy)

Nuanced choices within families

Weighted multilaterals (e.g., GEKS-Fisher vs GEKS-Törnqvist vs QU-GK) often give very similar results

Bigger impact:

- Bilateral vs multilateral
- Weights vs no weights
- Window length & splicing

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Evaluation approaches

Theoretical approaches:

- Evaluate index formulas against desirable mathematical and economic properties
- Include *axiomatic*, *economic* and *stochastic* approaches

Empirical approaches

- Evaluate index performance using real or simulated data
- Focus on *practical outcomes,* such as stability, sensitivity and interpretability

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Pilot, compare, decide

Empirical approach

- Test multiple candidate methods on the same dataset
- Compare:
  - Levels and growth rates
  - Match rates / drift
  - Volatility
- Document findings and rationale

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Practical considerations

Data characteristics

- Availability of weights
- Length and frequency of historic time series
- Product churn, missing data and classification consistency

Resources and systems

- Computational capacity and software tools
- Staff expertise and training needs
- Ability to implement and maintain chosen methods

Statistical policy and continuity

- Revision policy
- Need for transparency, reproducibility and international comparability
- Alignment with national CPI framework and user expectations

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Conclusions

Points covered:

- Introduction of elementary aggregate indices
- Discussion of the concepts of static and dynamic samples.
- Introduction of two main families of elementary indices considered for calculation of indices from alternative data sources: bilaterals and multilaterals.

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